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Towards improving the performance of traffic sign recognition using support vector machine based deep learning model

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Abstract
Nowadays autonomous vehicles are evolving due to the advancements in cutting edge technologies. In order to recognize the traffic signatures with high efficacy, traffic recognition system is required. Sign detection and classification are the two parts of the recognition system. The sign detection algorithm detects the size and coordinates of the sign board in an image and in sign classification, the representation of traffic signal is identified and classified into one of their traffic sign sub-classes. In order to achieve these goals, an extremely fast detection module using support vector machine is proposed to detect the traffic sign into one of the traffic classes such as, prohibitory, danger, mandatory, and non-sign. Further classification is carried out using deep convolutional neural networks to determine the sub-classes of each super-class, such as, prohibitory, danger, and mandatory. Based on publicly available benchmark traffic sign image datasets, we have demonstrated that the proposed approach has significantly improved traffic sign recognition accuracy compared with state-of-the-art systems.

Keywords Deep convolutional neural network · Maximally stable extremal region · Support vector machine · Traffic sign classification · Traffic sign detection

1 Introduction

Nowadays due to the advancement in technology, traffic sign recognition is popular in autonomous vehicles and driver assistant systems. The World Health Organization global status report on road safety reported that "Approximately 1.3 million people die each year on the world's roads, and between 20 and 50 million sustain non-fatal injuries" [32]. Therefore, it is on the urge to develop efficient traffic sign recognition systems. A

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typical traffic sign recognition system performs two tasks: traffic sign detection and traffic sign classification [20]. The traffic recognition system works through a camera that is mounted on the vehicle in the interior rear view mirror of the vehicles. This system generally captures and detects the traffic sign that is located at the left, right, or over the road and compares the traffic sign images available in the internal database [11]. Once the traffic sign is recognized, the current situation is notified to the driver through a visual interface which is available in the global positioning system. This automated system assists the drivers in an efficient manner as the drivers do not manually check the traffic signs [16].

Recently, machine learning techniques are used to recognize the traffic signs. The machine learning techniques are broadly categorized into supervised learning and unsupervised learning [30]. The traffic sign datasets consist of ground truths, so this problem comes under the category of supervised learning. The supervised learning algorithms such as decision tree, multi-layer perceptron, naive Bayes classifier, support vector machine (SVM), etc., have been used for traffic sign detection [36]. The proposed approach utilizes SVM for traffic sign detection as this technique finds the clear margin that separates the different categories of traffic signs between classes [35]. Furthermore, the traffic images are preprocessed and maximally stable extremal regions (MSER) are detected in order to improve the performance of the traffic recognition system.

Researchers in the area of machine learning use the term deep learning as a buzzword. A variety of deep learning techniques, such as convolutional neural networks (CNNs), auto-encoders, long short-term memories, etc., have been used for object detection and recognition, and they have produced better results than existing conventional techniques [28]. This has attracted the attention of the autonomous vehicles and driver assistant systems. The intuition behind utilizing deep learning, especially deep CNN (DCNN) in this study for traffic sign classification is that automatic feature extraction of traffic signs happens without human intervention [26]. The DCNN is a neural network model consists of two main parts: (i) alternating convolution and pooling and (ii) fully connected neural network with soft-max classification [35], [21]. Therefore, the layers in the DCNN, viz., convolutional layer, pooling layer, flatten layer, and fully connected layer are combined together to classify the different sub-classes of the detected traffic signs. The flatten layer converts the matrix representation of the feature map to a feature vector which is suitable for the learning process in the fully connected network. The reason behind traffic detection module before the classification module is to improve the performance of the traffic recognition system by reducing the false alarms. It is to be noted that higher the false alarms in the detection module, lower is the classifier accuracy in the classification module [20].

The key contributions of this study are as follows:

1. A performance improved traffic sign detection approach is proposed by extracting the MSER of the traffic sign image.
2. The structure of DCNN to classify the sub-classes of the three categories of traffic signs.
3. Performance analysis of the traffic sign recognition system.

Following the introduction, Section 2 covers related works. Section 3 details the proposed system. Section 4 discusses the results and performance of the proposed approach. Lastly, Section 5 summarizes the study and suggests additional improvements.

2 Related works

The reviews of traffic recognition methods were given in [20, 22]. The review of the existing traffic sign recognition systems with methodology, datasets used, evaluation metrics considered, advantages and disadvantages are tabulated in Table 1. The major subdivisions of traffic recognition, i.e., traffic sign detection and classification are very important nowadays in modern driving systems and unmanned autonomous vehicles. Extensive research has been done in the detection and classification of traffic road signs. The related works section is subdivided into two subsections, viz., detection of traffic signatures and classification of traffic signatures.

2.1 Detection of traffic signatures

The main objective of traffic sign detection is to realize the location and size of traffic signs present in the natural scene images. Figure 1 depicts the categories of traffic sign detection methods. The traffic sign detection techniques are classified into color-based method, shape-based method, and learning-based method. The specific method is chosen based on the problem and the system requirements, i.e., color-based methods are used for high resolution datasets [33]. In the color-based methods, the traffic signs have specific colors like red, blue, and yellow. The color characteristics are sensitive to the age of the traffic signs and the light variation in the traffic signs. Researchers work on various colour space models such as, RGB space, HSV space, HSI space, YUV space, etc., in order to improve the performance of the traffic sign detection systems [20]. The shape-based methods are based on detecting the edge of the images that are analyzed using structural approaches, i.e., geometric form of sign, closed contour, radial symmetry, distance of similarity, etc. These methods face the lacunae of changes in light, occlusions, changes in scale, rotations, and translations. In order to overcome these difficulties, learning-based detection methods are preferred [4].

In [5], authors used Adaboost learning algorithm to detect the traffic signs by cascading different Viola-Jones detector as classifiers. This approach suffered from high false positive rates. Also, this approach required large number of training images for performance improvement of the detectors. Authors in [6] used Adaboost classifier that is based on Viola-Jones detector and support vector regression. This approach was robust but the performance was not up to the mark in terms of accuracy. As the accuracy is one of the important performance metrics of traffic sign detection, there is still scope for improvement [24].

In [31], authors used linear discriminant analysis with SVM and they achieved high recall and precision. Furthermore, this approach was robust to bad lighting condition and partial occlusion. But this approach increased the computational cost of the detection module. Comparing to the above approaches, the proposed approach differs by combining the advantages of MSER and SVM inorder to detect the traffic signs with high accuracy.

2.2 Classification of traffic signatures

As part of traffic sign classification, all detected traffic signs are divided into sub-classes [2]. The traffic sign classification methods are categorized into hand-crafted features based classification and deep learning methods as depicted in Fig. 2. The hand-crafted features include Histogram of Oriented Gradients (HOG), speed up robust feature descriptor, binary robust invariant scalable key, local binary pattern, etc. In [7], authors in their work used

Table 1 Review of existing traffic sign recognition systems

Ref.	Methodology	Dataset used	Evaluation metrics	Advantanges	Disadvantages
[7]	Random forests	GTSRB, GTSDB, and Swedish traffic signs	Recall, Precision, Correct classification rate	High correct classification rate	Large search space
[19]	CNN	GTSRB	Accuracy	Small inter-class variance	High complexity
[1]	Deep CNN	Saudi Arabia traffic signs	Accuracy	High accuracy	Small dataset size leading to bias in the results
[39]	Optimized Lenet-5	GTSRB	Accuracy, Running time	High accuracy	High learning time
[9]	SVM and CNN	GTSRB and GTSDB	Recognition rate, Processing time	Less complexity	High misclassification rate
[8]	Semi-supervised classification	GTSRB	Accuracy, Recall, Precision, F1-Score	Computationally cheap	Inefficient labelling
[29]	CNN and computer vision	GTSRB	Accuracy	High accuracy	Low real time performance
[13]	Neural network	GTSRB, GTSDB, and Swedish traffic signs	Computing time, Precision, Recall, F-measure, Accuracy	High accuracy and low running time	Less robustness
[40]	Deep learning	Synthetic dataset	Recall, Precision, Accuracy	High accuracy	Unable to recognize multiple signs

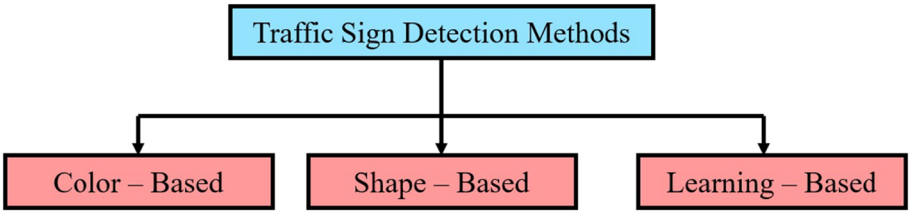


Fig. 1 Categories of traffic sign detection methods

different HOG descriptors to evaluate the performance of the ensemble random forest classifier. The performance of the classifier is unstable and the feature extraction process is time consuming. In [17], authors in their work used variety of hand-crafted features to classify the traffic sign but always the performance of the classifier depends on the segmentation process applied. In general, the hand-crafted features face the lacuna of domain expert dependency [3].

The deep learning approaches overcome the disadvantage of the hand-crafted features based classification method. In [23] and [38], authors used CNN to classify the traffic sign. They obtained high accuracy but took huge training time without GPU implementation. In [23] and [12], the authors used ensemble of CNNs and obtained better performance but the computations involved in the classification process is very complex. Comparing the works discussed in the literature, the proposed classifier is different in which the DCNN consists of two levels of convolutional and pooling layers, then a flattened layer to convert the transformed image feature matrix to feature vector, and fully connected layer with different number of nodes in the output layer which depends on the category of traffic class.

3 Proposed approach

The proposed traffic sign recognition system consists of the following modules: Noise removal, Double normalization, MSER detection, Traffic sign detection, and Traffic sign classification. Figure 3 depicts the flow diagram of the proposed approach. The proposed approach takes the natural scene image as input to detect and classify the traffic sign in the image.

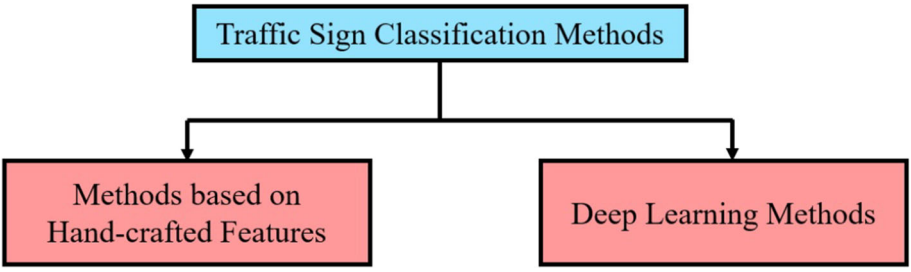


Fig. 2 Categories of traffic sign classification methods

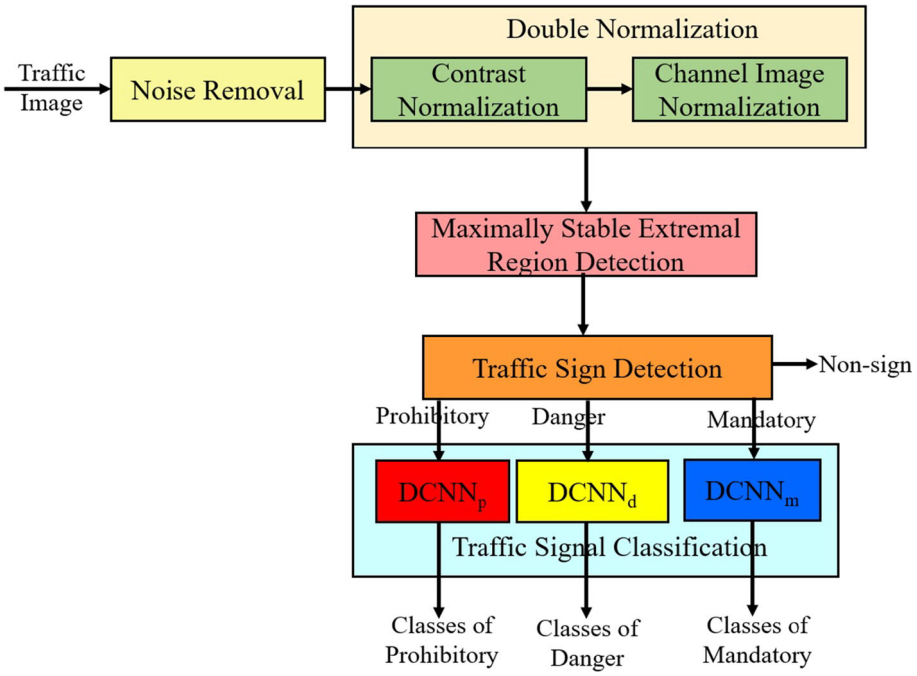


Fig. 3 Flow diagram of the proposed approach

3.1 Noise removal

The real-time images consist of huge noise on all the channels. The salt and pepper noise in the image is removed using median blur filter [36]. This filter computes the median of the neighborhood pixels. Furthermore, this filter could not create new value for the pixel and therefore any unrepresentative pixel in the neighborhood will not affect the result. This means that the filter preserves the edges and round corners. This module deletes the scratches present in the image. The median blur filtered image, $mbf(a, b)$ is obtained from the median pixel values in the neighborhood of (a, b) in the input traffic sign image $t(a, b)$ as follows:

$$mbf\ (a, b) = median\left(\sum_{i=-1}^{1}\sum_{j=-1}^{1} t\ (a-i, b-j)\right) \tag{1}$$

Figure 4 illustrates the computation of median blur filter. The sample filter size is 3 x 3. The 3 x 3 matrix is converted to a vector and the vector is sorted to ascending order and the median of the pixel value is replaced in the original image to obtain the median filtered image.

3.2 Double normalization

Double normalization is applied to the filtered image in order to make the image suitable for further process. The following are the steps to perform double normalization:

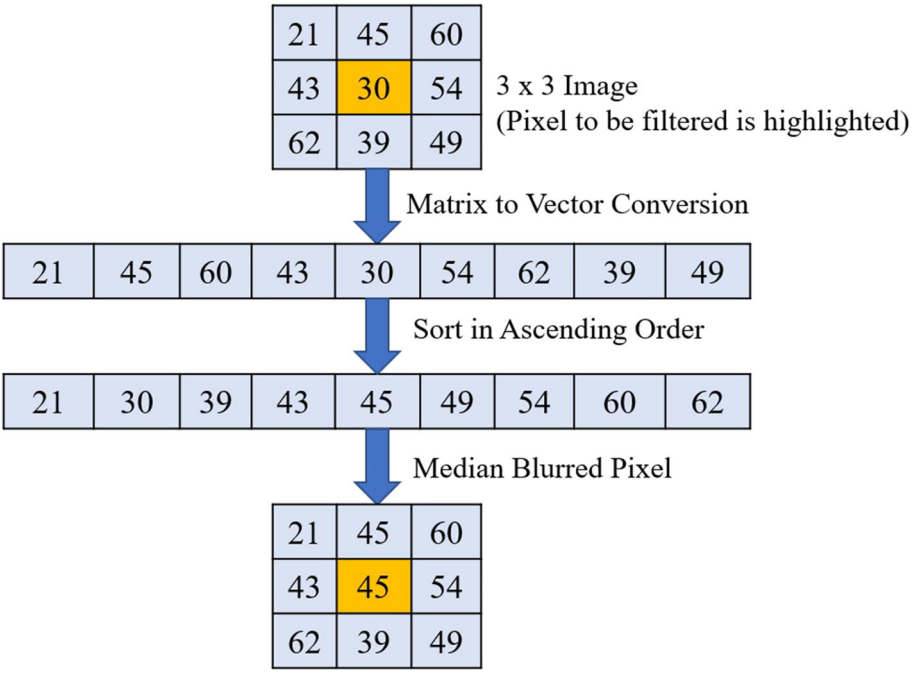


Fig. 4 Illustration of noise removal

3.2.1 Contrast normalization

Each channel of the RGB image is normalized to the range of [0,1]. The contrast normalization is applied to normalize the image and is performed as follows:

$$Diff = Channel_{max} - Channel_{min} \tag{2}$$

where $Channel_{max}$is the maximum value of the channel and $Channel_{min}$is the minimum value of the channel. Then the computed difference is used to perform contrast normalization.

$$Channel_{CN} = \frac{1}{Diff} \times (Channel_i - Channel_{min}) \tag{3}$$

where $Channel_i$is the channel to be normalized.

3.2.2 Channel image normalization

After performing contrast normalization, channel image normalization is applied for the blue and red channel. Eq. 3 is utilized for normalizing each channel separately.

3.3 Maximally stable extremal region detection

A minimum and maximum area of 32 x 32 and 10k respectively are chosen for MSER detection to avoid too much noise in small areas. As soon as the red and blue MSER regions are

detected, a binary image is created for both regions. In order to model the colour composition of a traffic sign, the HSV colour space can better identify hue, saturation, and value bands. Furthermore, it isolates intensity/brightness, which aids in robustness to illumination, instead of RGB. These ranges are extracted by choosing colour ranges in the HSV colour space. When these ranges of colours are chosen, all other colours in the image are deleted, and the image becomes a binary image. It can only be detected in the red and blue range of saturation and lightness. It is to be noted that we normalized the HSV image between the range [0, 1].

To eliminate false results, the common part of the image in which the HSV threshold image and the MSER generated image are considered and filled the inside area of the connected region is considered. As there is no common part in HSV and MSER, the red channel in the following image becomes zero. This method removes more noise and improves detection accuracy. On the combined image of HSV and MSER, the blob or contours is applied. If the contour area is greater than 300, the image is cropped to avoid false readings. Between 1.21 and 0.59, the aspect ratio filter is used. If the aspect ratio of the object matches, the image will be forwarded and cropped on that basis.

3.4 Traffic sign detection

In order to detect the traffic sign, the SVM model with Radial Basis Function (RBF) kernel is trained with the HOG feature of image. The intuition behind utilizing SVM for traffic sign detection is that it is a powerful mechanism for multi-class classification as it has the tendency that not to overfit in any circumstance. SVM classifiers use statistical learning theory to create a hyperplane to separate classes of positive instances from those of negative instances based on statistical learning theory. Hyperplanes are used in SVM to classify data points. SVM maximises support vector margins between classes since there is a need to separate all classes. The main objective of SVM classification is to find the hyper-plane in the high dimensional feature space that separates the data points to their respective classes. The data is transformed to infinite dimensional space, S using the mapping function ϕ and it is represented as $\phi(g_i)^T \phi(g_j)$. If $g = (g_1, g_2)$ and $\phi(x) = \left(g_1^2, g_2^2, \sqrt{2g_1g_2}\right)$ then the kernel function is defined as follows:

$$K(g_i, g_j) = \phi(g_i)^T \phi(g_j) \tag{4}$$

The SVM is designed in such a way that to compute $\alpha_1, \alpha_2, ..., \alpha_N$ in which $Q(a)$ is to be maximized. This is computed as follows:

$$Q(a) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{N} \alpha_i \alpha_j y_i y_j K(g_i, g_j) \tag{5}$$

The constraint to be followed with respect to α is as follows:

$$\sum_{i=1}^{N} \alpha_i y_i = 0, C \geq \alpha_i \geq 0, \forall \alpha_i \tag{6}$$

The classification is performed as follows:

$$g(x) = \sum \alpha_i y_i K(g_i, g_j) + b \tag{7}$$

The RBF kernel function is utilized as a kernel trick in the proposed approach which is computed as follows:

$$RBF_k(r_i, s_j) = exp(-\gamma ||r_i - s_j||^2) \tag{8}$$

where r_i and s_j are the data points. This function calculates the distance score between these two points. It is to be noted, that higher the score, closer is the data points. The kernel trick reduces the computational time where huge amounts of data are used. Furthermore, it prevents complex transformation.

The classification is performed by separating the points in the feature space. When the feature points are separated, the distance of the hyperplane is computed by solving the following optimization task:

$$\phi(w) = \frac{1}{2} \parallel w \parallel^T + \sum_{i=1}^{N} (\xi_i - \rho\nu) \tag{9}$$

The objective is to minimize this optimization task such that the following condition is satisfied.

$$w.\phi(x_i) \geq \rho - \xi_i, \xi_i \geq 0 \tag{10}$$

where ξ_i are slack variables and ρ is offset from origin. For multi-class classification, one-versus-one approach is used. The number of classifiers are retrieved using the following equation:

$$N_{classifier} = n \times (n-1)/2 \tag{11}$$

where n is the number of classes. Here, each classifier separates points of two different classes and combining all the one-versus-one classifiers lead to the multi-class classifier. Then the trained model on image differentiates sign and non-sign contours and save non-sign contours image in one directory. Again the SVM model is trained with RBF kernel and image HOG feature to divide into four sub-classes: prohibitory sign (circular, white ground with red border), mandatory sign (circular, blue ground), danger sign (triangular, yellow ground with black border), and non-sign. Then these classes are passed to the traffic signal classification module except non-sign to classify the sub-classes of all the three traffic classes.

3.5 Traffic signal classification

Traffic signal classification classifies the detected traffic sign to their sub-classes. Figures 5, 6, and 7 depict the sub-classes of the three traffic classes, prohibitory, danger, and mandatory respectively. Traffic signal classification is performed by resizing all images. The computational efficacy of the model is improved by training three DCNNs with the structure as depicted in Fig. 8. Each model is trained with different classes of traffic sign,



Fig. 5 Sub-classes of prohibitory



Fig. 6 Sub-classes of danger



Fig. 7 Sub-classes of mandatory

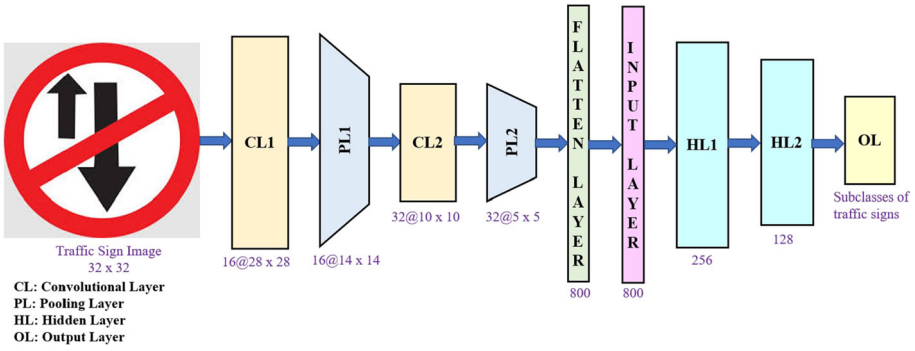


Fig. 8 Structure of deep convolutional neural network

viz., prohibitory, danger and mandatory to classify their sub-classes. The DCNN consists of convolution layers, pooling layers, flatten layer, and fully connected multi-layer perceptron.

Let the size of the input traffic sign image be CL_{in} X CL_{in} X Z_{in} and Z_{out} be the number of kernels with the spatial size S_s and stride S_L. The padding is denoted as CL_p and the size of the transformed image output is computed as follows:

$$CL_{out} = \frac{CL_{in} - S_s + 2C_p}{S_t} + 1 \tag{12}$$

The size of the transformed image is CL_{out} X CL_{out} X Z_{out}. The convolution layer automatically extracts the features and the transformed features are passed to the pooling layer. The pooling layer reduces the spatial size of the traffic sign image. Max pooling is used for pooling and is computed as follows:

$$PL(CL_{ij}^{I}, CL_{ik}^{I}) = max(CL_{ij}^{I}, CL_{ik}^{I}) \tag{13}$$

After performing two levels of convolution and pooling, flattened layer is used to convert the transformed image to a vector which is given as input to the fully connected network. The input to the Hidden Layer (HL) is computed by multiplying the corresponding weights, W^{HL}, with the feature vector as follows:

$$I_j^{HL_i} = \sum_{i=1}^{p} PL_i \times W_{ij}^{HL_i} \tag{14}$$

In the fully connected network, Rectified Linear Unit (ReLU) is used in the hidden layers and the output of the HL is computed as follows:

$$f\left(I_j^{HL_{i+1}}\right) = max\left(I_j^{HL_i}, 0\right) \tag{15}$$

As a result of multiplying the corresponding weights with the output of the last HL, and adding a bias, the output of a fully connected network is computed as follows:

$$O_i^c = \sum_{i=1}^{p} HL_i \times W_{ij}^{HL} + Bias \tag{16}$$

The softmax activation function is used in the output layer to classify the sub-classes of each of the three traffic classes as the traffic classes have multiple sub-classes and is computed as follows:

$$SM\left(O_{ij}\right) = \frac{Exp\left(O_{ij}\right)}{\sum_{j=0}^{k} Exp\left(O_{ij}\right)} \tag{17}$$

The number of nodes in the output layer varies depending on the number of sub-classes in each of the traffic sign class.

4 Results and discussions

The proposed approach was implemented in Python with 16 GB RAM machine. Experiments were conducted using German Traffic Sign Detection Benchmark (GTSDB) and German Traffic Sign Recognition Benchmark (GTSRB) datasets to test the efficacy of the proposed traffic sign detection and classification system [10, 27]. There are 900 images in the GTSDB, of which 600 were used for training and 300 for testing. This dataset was used

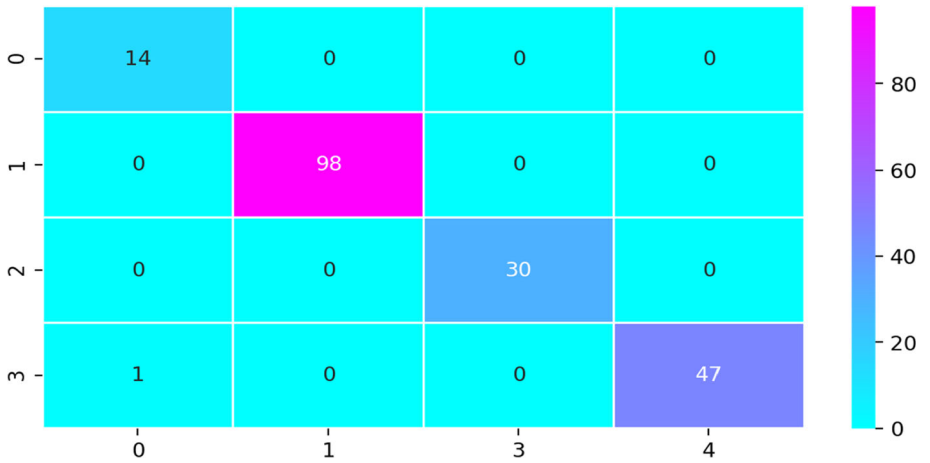


Fig. 9 Confusion matrix of traffic sign detection

for single image detection problem. The GTSRB is a multi-class single image classification dataset. This dataset consists of 43 classes with 51,840 images in which 39,210 images were used for training and 12,630 were used for testing.

4.1 Performance analysis of the proposed approach

The performance of the proposed approach was evaluated based on the the performance metrics, accuracy and loss rate which are computed as follows:

$$Accuracy = \frac{Number\ of\ correctly\ detected\ classes}{Total\ number\ of\ classes} \tag{18}$$

$$Loss\ Rate = \frac{Number\ of\ wrongly\ detected\ classes}{Total\ number\ of\ classes} \tag{19}$$

The traffic sign is detected and the confusion matrix for the detection module is depicted in Fig. 9. The detection module detects the traffic sign into four categories, viz., prohibitory, danger, mandatory, and non-sign. After detecting the traffic signs, the prohibitory, danger, and mandatory traffic signs are classified into their sub-classes. The accuracy obtained using the proposed approach for traffic sign detection is 99.47%. The sub-classes of each of the traffic categories are tabulated in Table 2.

Table 2 Sub-categories of the traffic sign classes

Traffic sign classes	Sub-classes	Number of sub-classes
Prohibitory	–1, 0, 1, 2, 3, 4, 5, 7, 8, 9, 10, 15, 16	13
Danger	–1, 11, 18, 19, 20, 21, 22, 23, 24, 25 26, 27, 28, 29, 30, 31	16
Mandatory	–1, 33, 34, 35, 36, 37, 38, 39, 40	9

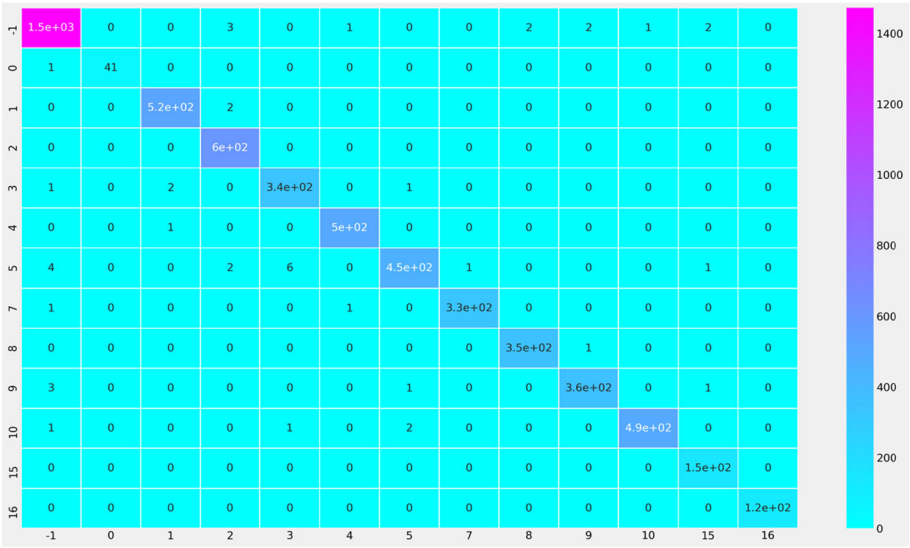
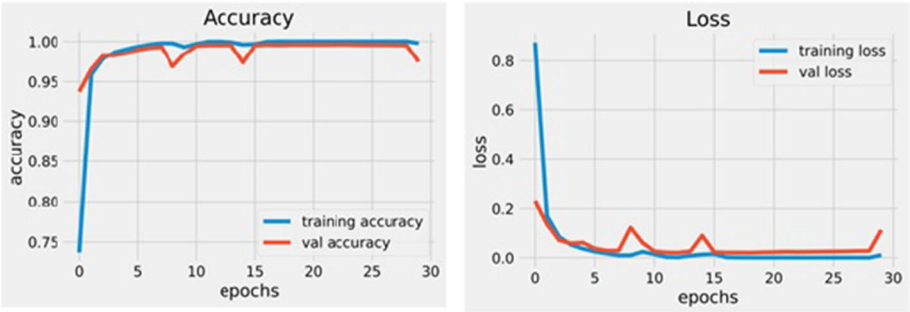


Fig. 10 Confusion matrix of prohibitory traffic class

Figure 10 depicts the confusion matrix of the prohibitory traffic class. Figure 11(a) and (b) depicts the accuracy and loss rate respectively of the prohibitory traffic class for both training and validation. The proposed structure is trained for 30 epochs and stopped as there is no improvement behind these epochs. The accuracy obtained using the proposed $DCNN_p$ structure for traffic sign classification is 97.41%.

Figure 12 depicts the confusion matrix of the danger traffic class. Figure 13(a) and (b) depicts the accuracy and loss rate respectively of the danger traffic class for both training and validation. The proposed structure is trained for 30 epochs and stopped as there is no improvement behind these epochs. The accuracy obtained using the proposed $DCNN_d$ structure for traffic sign classification is 98.94%.



(a) (b)

Fig. 11 Performance analysis of prohibitory traffic class (a) Accuracy (b) Loss Rate

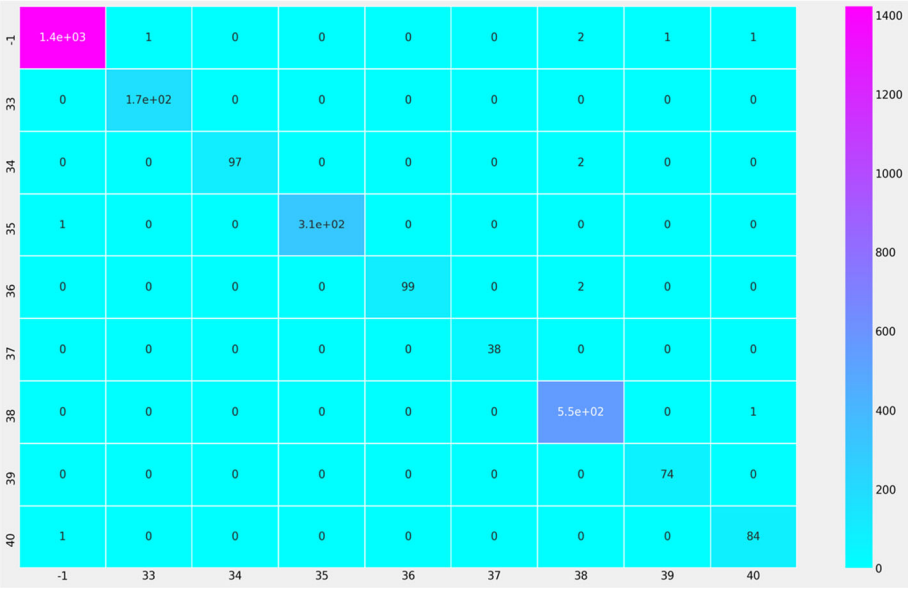
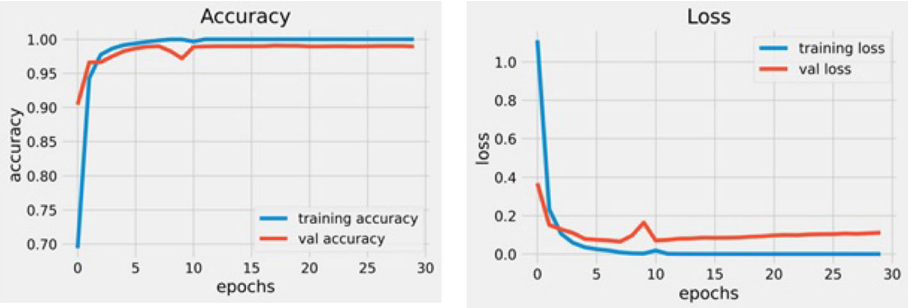


Fig. 12 Confusion matrix of danger traffic class

Figure 14 depicts the confusion matrix of the mandatory traffic class. Figure 15(a) and (b) depicts the accuracy and loss rate respectively of the mandatory traffic class for both training and validation. The proposed structure is trained for 30 epochs and stopped as there is no improvement behind these epochs. The accuracy obtained using the proposed $DCNN_m$ structure for traffic sign classification is 99.47%.

Figure 16 depicts six sample test traffic scene images. Figure 17(a), (b), (c), and (d) depict the contrast normalization of red colour, contrast normalization of blue color, threshold in HSV with red color, and threshold in HSV with blue color respectively for all the six



(a) (b)

Fig. 13 Performance analysis of danger traffic class (a) Accuracy (b) Loss Rate

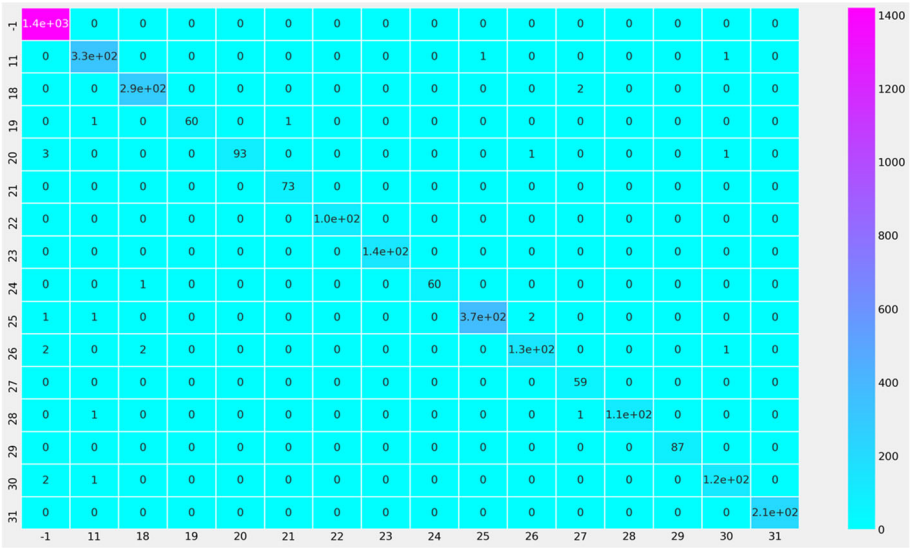
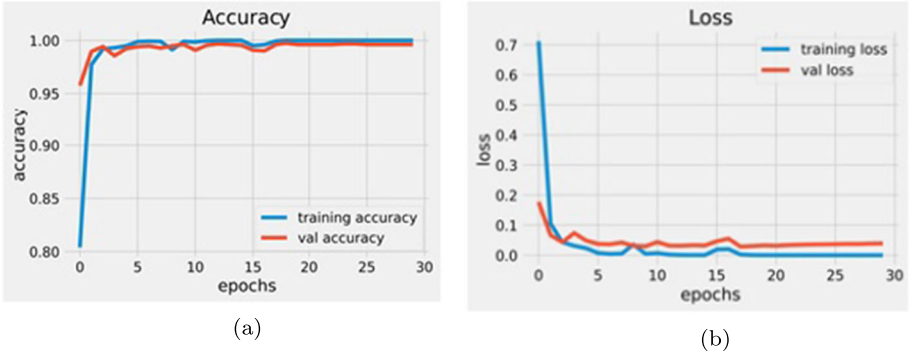


Fig. 14 Confusion matrix of mandatory traffic class

test images. Figures 18 and 19 depict the contour images and traffic sign detection of the test images respectively. It is observed that the proposed approach detects the traffic sign in road traffic scenery with high efficacy.

4.2 Comparison with state-of-the art traffic sign recognition approaches

A comparison is made between the proposed approach and the state-of-the-art traffic sign recognition system. Table 3 compares the performances of the proposed approach and existing traffic sign recognition systems in terms of detection and classification. In all



(a) (b)

Fig. 15 Performance analysis of mandatory traffic class (a) Accuracy (b) Loss Rate



Fig. 16 Original test images (a) Test Image 1 (b) Test Image 2 (c) Test Image 3 (d) Test Image 4 (e) Test Image 5 (f) Test Image 6

the existing traffic sign recognition systems used for comparison utilized GTSDB and GTSRB datasets for detection and classification respectively. The proposed method performs double normalization and extracted MSER before detection and classification, which is evidently superior to the existing approaches. Table 4 tabulates the performance comparison of proposed approach in terms of three traffic sign classes, viz., prohibitory, danger, and mandatory. The proposed approach outperforms the existing approaches in mandatory classes and significantly shown performance in other classes.

5 Conclusion

A SVM-based deep learning model is proposed in this study to improve the performance of traffic sign detection and classification. The noise in the traffic sign image is removed and pre-processed using double normalization. Then the maximally stable extremal region



Fig. 17 (a) Contrast Normalization of red colour (b) Contrast Normalization of blue colour (c) Threshold in HSV colour space with red colour (d) Threshold in HSV colour space with blue colour



Fig. 18 Contour Images (a) Test Image 1 (b) Test Image 2 (c) Test Image 3 (d) Test Image 4 (e) Test Image 5 (f) Test Image 6



Fig. 19 Detection of Traffic Sign Images (a) Test Image 1 (b) Test Image 2 (c) Test Image 3 (d) Test Image 4 (e) Test Image 5 (f) Test Image 6

Table 3 Comparison of proposed approach with state-of-the-art approaches

Ref./Year	Detection (%)	Classification (%)
[15], 2014	96.95	–
[36], 2015	97.72	97.75
[3], 2016	97.04	–
[19], 2016	98.86	–
[34], 2016	97.94	–
[11], 2017	92.63	80.5
[25], 2018	99.43	–
[4], 2019	96.85	96.23
[9], 2020	99.37	–
[8], 2020	98.59	–
[40], 2022	–	97.70
Proposed	**99.47**	**98.65**

Bold emphasis represents high performance

is extracted in order to enhance the performance of traffic sign detection. Also, the traffic sign is detected using multi-class SVM. Finally, the detected traffic signs are classified into their subclasses using DCNN. Experiments were conducted on benchmark traffic sign image datasets and analyzed with respect to two performance metrics, viz., accuracy and loss rate. Furthermore, it is crystal clear that the proposed approach outperforms the state-of-the-art traffic sign recognition systems in terms of both detection and classification. This study can be extended in future to recognize the traffic signs in real-time road traffic videos.

Table 4 Comparison of proposed approach with state-of-the-art classifications

Ref./Year	Prohibitory (%)	Danger (%)	Mandatory (%)
[31], 2013	99.88	99.85	99.85
[10], 2013	98.80	74.60	67.30
[18], 2013	100	100	96.98
[14], 2014	100	92.20	98.57
[6], 2015	100	100	99.87
[37], 2016	99.45	98.33	96.50
Proposed	97.41	98.94	**99.60**

Bold emphasis represents high performance

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Data Availability Data sharing not applicable to this article as no datasets were generated during the current study.

Declarations

Ethics approval None.

Informed consent None.

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